

Simon Oddershede Gregersen

Email gregersen@cispa.de
Website <https://simongregersen.com>
ORCID [0000-0001-6045-5232](https://orcid.org/0000-0001-6045-5232)
DBLP <https://dblp.org/pid/263/1059>

Education

2017 – 2023 **PhD in Computer Science**. Aarhus University.
Advised by Amin Timay and Lars Birkedal.
Thesis: Higher-Order Separation Logic for Distributed Systems and Security.
2017 – 2020 **MSc in Computer Science**. Aarhus University.
2014 – 2017 **BSc in Computer Science**. Aarhus University.

Employment

2026 – **CISPA Helmholtz Center for Information Security**. Tenure-track faculty.
2024 – 2025 **New York University**. Postdoctoral fellow.
2023 – 2024 **Aarhus University**. Postdoctoral researcher.
2021 **Huawei Technologies R&D (UK), Edinburgh**. Research intern.
2017 **Aarhus University**. Center for Advanced Software Analysis. Student programmer.
2015 – 2017 **CCI Europe A/S (now Stibo DX)**. Student software developer.

Publications

- [13] Philipp G. Haselwarter, Alejandro Aguirre, Simon Oddershede Gregersen, Joseph Tassarotti, and Lars Birkedal. “Modular Verification of Differential Privacy in Probabilistic Higher-Order Separation Logic”. In: *Proc. ACM Program. Lang.* 10.PLDI (2026). To appear.
- [12] Simon Oddershede Gregersen, Chaitanya Agarwal, and Joseph Tassarotti. “Logical Relations for Formally Verified Authenticated Data Structures”. In: *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security, CCS 2025, Taipei, Taiwan, October 13-17, 2025*. 2025, pp. 1394–1408. DOI: [10.1145/3719027.3744801](https://doi.org/10.1145/3719027.3744801).
- [11] Kwing Hei Li, Alejandro Aguirre, Simon Oddershede Gregersen, Philipp G. Haselwarter, Joseph Tassarotti, and Lars Birkedal. “Modular Reasoning about Error Bounds for Concurrent Probabilistic Programs”. In: *Proc. ACM Program. Lang.* 9.ICFP (2025). DOI: [10.1145/3747514](https://doi.org/10.1145/3747514).
- [10] Philipp G. Haselwarter, Kwing Hei Li, Alejandro Aguirre, Simon Oddershede Gregersen, Joseph Tassarotti, and Lars Birkedal. “Approximate Relational Reasoning for Higher-Order Probabilistic Programs”. In: *Proc. ACM Program. Lang.* 9.POPL (2025). DOI: [10.1145/3704877](https://doi.org/10.1145/3704877).
- [9] Philipp G. Haselwarter, Kwing Hei Li, Markus de Medeiros, Simon Oddershede Gregersen, Alejandro Aguirre, Joseph Tassarotti, and Lars Birkedal. “Tachis: Higher-Order Separation Logic with Credits for Expected Costs”. In: *Proc. ACM Program. Lang.* 8.OOPSLA2 (2024). DOI: [10.1145/3689753](https://doi.org/10.1145/3689753).

- [8] Simon Oddershede Gregersen, Alejandro Aguirre, Philipp G. Haselwarter, Joseph Tassarotti, and Lars Birkedal. “Almost-Sure Termination by Guarded Refinement”. In: *Proc. ACM Program. Lang.* 8.ICFP (2024). DOI: [10.1145/3674632](https://doi.org/10.1145/3674632).
- [7] Alejandro Aguirre, Philipp G. Haselwarter, Markus de Medeiros, Kwing Hei Li, Simon Oddershede Gregersen, Joseph Tassarotti, and Lars Birkedal. “Error Credits: Resourceful Reasoning about Error Bounds for Higher-Order Probabilistic Programs”. In: *Proc. ACM Program. Lang.* 8.ICFP (2024). DOI: [10.1145/3674635](https://doi.org/10.1145/3674635).
- [6] Amin Timany, Simon Oddershede Gregersen, Léo Stefanescu, Jonas Kastberg Hinrichsen, Léon Gondelman, Abel Nieto, and Lars Birkedal. “Trillium: Higher-Order Concurrent and Distributed Separation Logic for Intensional Refinement”. In: *Proc. ACM Program. Lang.* 8.POPL (2024). DOI: [10.1145/3632851](https://doi.org/10.1145/3632851).
- [5] Simon Oddershede Gregersen, Alejandro Aguirre, Philipp G. Haselwarter, Joseph Tassarotti, and Lars Birkedal. “Asynchronous Probabilistic Couplings in Higher-Order Separation Logic”. In: *Proc. ACM Program. Lang.* 8.POPL (2024). DOI: [10.1145/3632868](https://doi.org/10.1145/3632868).
- [4] Simon Oddershede Gregersen, Johan Bay, Amin Timany, and Lars Birkedal. “Mechanized Logical Relations for Termination-Insensitive Noninterference”. In: *Proc. ACM Program. Lang.* 5.POPL (2021). DOI: [10.1145/3434291](https://doi.org/10.1145/3434291).
- [3] Léon Gondelman, Simon Oddershede Gregersen, Abel Nieto, Amin Timany, and Lars Birkedal. “Distributed Causal Memory: Modular Specification and Verification in Higher-Order Distributed Separation Logic”. In: *Proc. ACM Program. Lang.* 5.POPL (2021). DOI: [10.1145/3434323](https://doi.org/10.1145/3434323).
- [2] Morten Krogh-Jespersen, Amin Timany, Marit Edna Ohlenbusch, Simon Oddershede Gregersen, and Lars Birkedal. “Aneris: A Mechanized Logic for Modular Reasoning about Distributed Systems”. In: *Programming Languages and Systems - 29th European Symposium on Programming, ESOP 2020, Held as Part of the European Joint Conferences on Theory and Practice of Software, ETAPS 2020, Dublin, Ireland, April 25-30, 2020, Proceedings*. 2020. DOI: [10.1007/978-3-030-44914-8_13](https://doi.org/10.1007/978-3-030-44914-8_13).
- [1] Simon Oddershede Gregersen, Søren Eller Thomsen, and Aslan Askarov. “A Dependently Typed Library for Static Information-Flow Control in Idris”. In: *Principles of Security and Trust - 8th International Conference, Held as Part of the European Joint Conferences on Theory and Practice of Software, ETAPS 2019, Prague, Czech Republic, April 6-11, 2019, Proceedings*. 2019. DOI: [10.1007/978-3-030-17138-4_3](https://doi.org/10.1007/978-3-030-17138-4_3).

Awards

- Distinguished Paper Award at ICFP 2024.
- Carlsberg Foundation Internationalization Fellowship ([CF23-0791](#)). DKK 1,020,000 ($\approx$ USD 150,000).

Academic Service

Organizing

- 2026 Artifact evaluation co-chair. [Object-Oriented Programming, Systems, Languages, and Applications \(OOPSLA\)](#) .

Program Committees

- 2027 Certified Programs and Proofs (CPP).
- 2026 [SPLASH Onward! Papers](#).
 [Recent Advances in Concurrency and Logic \(RADICAL\)](#).
 [Foundations of Computer Security \(FCS\)](#).
 [Programming Language Design and Implementation \(PLDI\)](#).
- 2025 [Recent Advances in Concurrency and Logic \(RADICAL\)](#).

External Reviewing

Journals JFP (2025), TCS (2024), TOSEM (2023)

Conferences LICS 2025, LICS 2024, CCS 2022, POPL 2022, OOPSLA 2022, ICFP 2022, CSF 2022, CSF 2021, ESOP 2020

Selected Talks

- 2026 Feb *Building Extensible Program Logics through Effect Handlers.*
MPI-SWS Foundations of Programming group seminar
- 2025 Oct *Logical Relations for Formally Verified Authenticated Data Structures.*
New England Systems Verification Day.
- 2025 Jun *Logical Relations for Formally Verified Authenticated Data Structures.*
The Fifth Iris Workshop.
- 2025 May *Logical Relations for Formally Verified Authenticated Data Structures.*
New Jersey Programming Languages and Systems Seminar.
- 2025 Apr *Logical Relations for Formally Verified Authenticated Data Structures.*
VU Amsterdam PLSec reading group.
- 2024 Apr *Trillium: Intensional Refinement in Higher-Order Separation Logic.*
New England Systems Verification Day.
- 2023 Jul *Asynchronous Probabilistic Couplings in Higher-Order Separation Logic.*
Bristol Programming Languages Research group seminar.
- 2023 Jul *Asynchronous Probabilistic Couplings in Higher-Order Separation Logic.*
Workshop on Verification of Probabilistic Programs (VeriProP).
- 2022 May *Trillium: History-Sensitive Refinement in Separation Logic.*
The Second Iris Workshop.
- 2020 Nov *Mechanized Logical Relations for Termination-Insensitive Noninterference.*
Chalmers ProgLog/Security seminar.